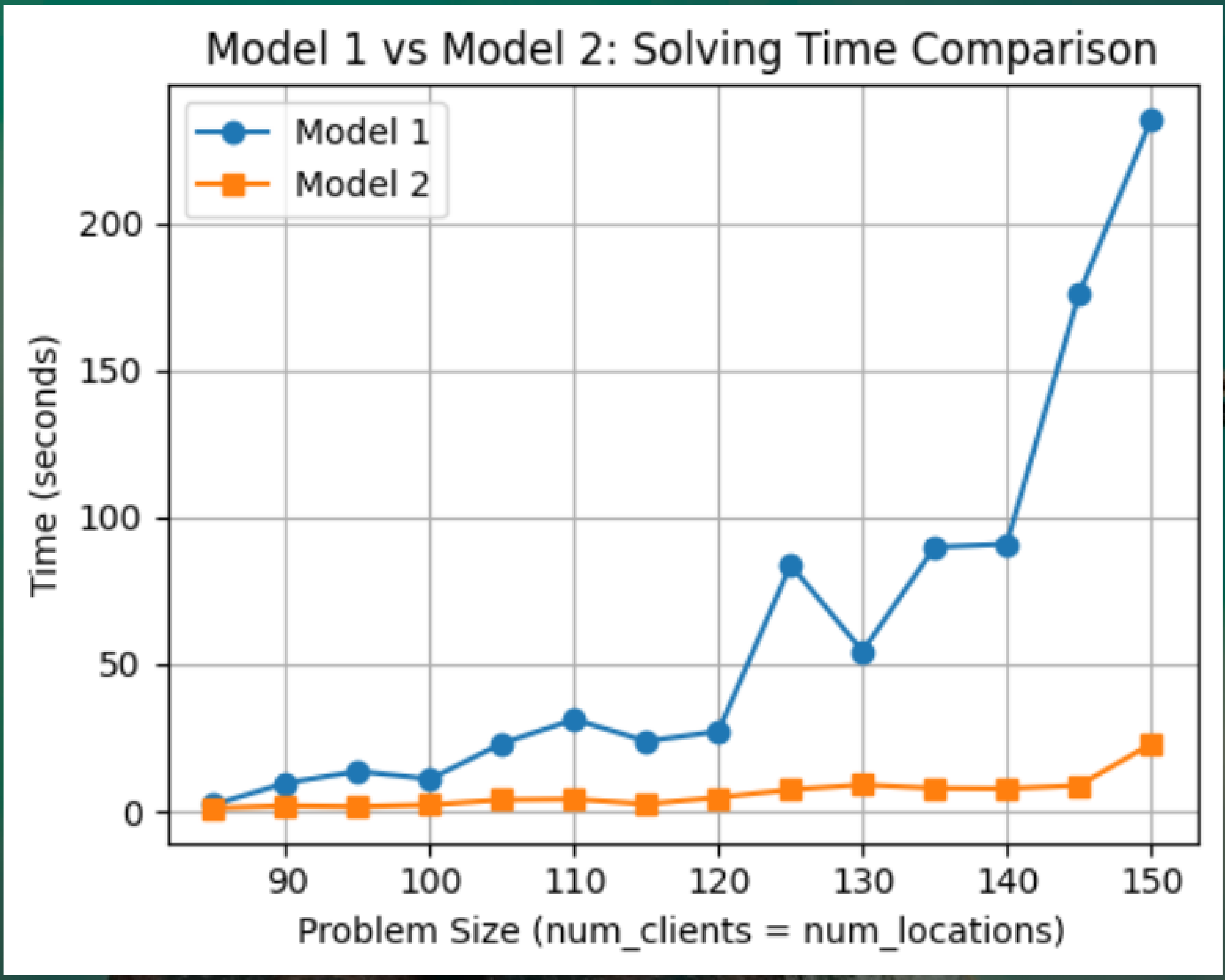
Know the trick = 10x better performance



The Art of Mathematical Modeling

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JPJ SOLUTIONS

Example

Maximum Coverage Problem

Objective:

- 🏪 Place facilities
- 👨 Cover as many clients as possible

Constraints:

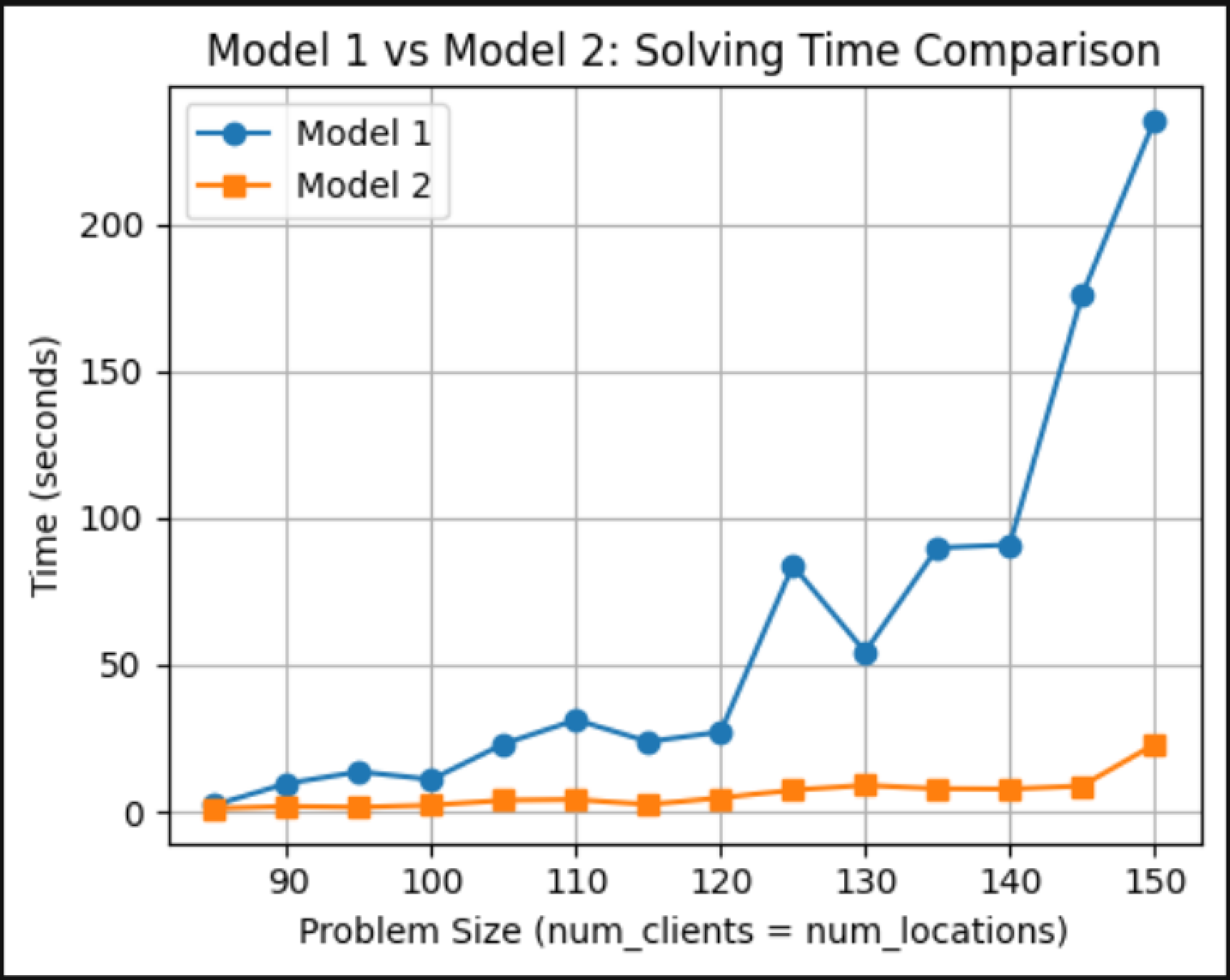
- 🧩 Every facility serves different clients
- 💰 You can only place B facilities in total

The catch

- 📝 You can formulate this problem in two different ways.
- ✅ Either, when solved, yields the optimal placement of facilities.
- 😏 But one formulation is solved much faster

Model comparison (1/2)

Model 2 runs more than 10x faster and scales much better.



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Model comparison (2/2)

	Model 1	Model 2
Variables	$O(n \cdot m)$	$O(n + m)$
Constraints	$O(n \cdot m)$	$O(m)$

where $n = |N|$ (number of facilities) and m (number of clients).

Live Session and more Modeling Know-How:

Registration Link in the comments!

JPJ
SOLUTIONS

Model 1

Decision Variables:

$$x_i = \begin{cases} 1 & \text{if facility is placed at location } i \in N \\ 0 & \text{otherwise} \end{cases}$$

$$y_{ij} = \begin{cases} 1 & \text{if facility } i \text{ serves client } j \\ 0 & \text{otherwise} \end{cases} \qquad i \in N,\ j \in S_i$$

Objective:

$$\max \sum_{i \in N} \sum_{j \in S_i} y_{ij}$$

Subject to:

$$\sum_{i \in N} x_i \leq B$$

$$\sum_{i: j \in S_i} y_{ij} \leq 1 \quad \forall j$$

$$y_{ij} \leq x_i \quad \forall i \in N,\ j \in S_i$$

where S_i is the set of clients that facility i can serve.

JPJ SOLUTIONS

Model 2

Decision Variables:

$$x_i = \begin{cases} 1 & \text{if facility is placed at location } i \in N \\ 0 & \text{otherwise} \end{cases}$$

$$z_j = \begin{cases} 1 & \text{if client } j \text{ is served} \\ 0 & \text{otherwise} \end{cases}$$

Objective:

$$\max \sum_j z_j$$

Subject to:

$$\sum_{i \in N} x_i \leq B$$

$$z_j \leq \sum_{i \in S_j} x_i \quad \forall j$$

where S_j is the set of facilities that can serve client j.